Part	Model	PPV	NPV	FPR	FNR
b.	Overall	0.81	0.79	0.14	0.28
b.	Group 0	0.83	0.82	0.12	0.23
b.	Group 1	0.75	0.71	0.17	0.40
c.	Overall	0.81	0.79	0.14	0.28
c.	Group 0	0.83	0.82	0.12	0.23
c.	Group 1	0.75	0.79	0.17	0.40
d.	G = 0 Overall	0.85	0.88	0.12	0.15
d.	G = 1 Overall	1.00	1.00	0.00	0.00

b. Look at the LogisticRegression model trained in `part_b()` and shown in the top left plot. In `free_response/5a.png`, the area shaded grey shows positive predictions; the area shaded red shows negative predictions.
*** To what extent does the classifier treat individuals in Group 0 differently from those in Group 1?**
*** If you were applying for a loan and this classifier were making the decision, would you rather be a member of Group 0 or Group 1? Why?**
*** When you look at the Overall, Group 0 and Group 1 metrics you added to the table above, which group's metrics are most similar to that of the Overall metrics? Why?**
The gray shaded area represents positive predictions, while the red shaded area represents negative predictions.

- The classifier treats individuals in Group 0 differently from those in Group 1 to a certain extent. The decision boundary of the classifier is shown in the top left plot. The logistic regression model's coefficients are 0.049 for income and 0.074 for credit score, which suggests that credit score is more important than income in predicting the loan status. The negative coefficient of -6.562 indicates that a lower credit score and income are associated with a higher likelihood of being denied a loan. Individuals in Group 0 tend to have lower incomes and credit scores, while individuals in Group 1 tend to have higher incomes and credit scores. As a result, the classifier may be more likely to predict that individuals in Group 1 are more likely to be approved for a loan than those in Group 0.
- If I were applying for a loan and this classifier was making the decision, I would rather be a member of Group 1 because it has higher positive predictive value (PPV) and negative predictive value (NPV) than Group 0. This means that if I were in Group 1, the

probability of being approved for a loan is higher, and the probability of being denied a loan is lower than if I were in Group 0.

- The metrics for Group 1 are more similar to those of the Overall metrics than those of Group 0. This is because the classifier is more likely to predict positive outcomes for individuals in Group 1, and Group 1 has a higher PPV and NPV than Group 0. However, the false negative rate (FNR) for Group 1 is higher than that of Group 0, indicating that the classifier is more likely to incorrectly predict negative outcomes for individuals in Group 1 who would have been approved for a loan.

c. Consider the LogisticRegression model trained in `part_c()` and shown in the top right plot of `free_response/5a.png`. Looking at the code for `part_b` and `part_c`, what's different about how this model was trained compared to the model from part __b.__? Does this model's decision boundary and fairness metrics differ from those of the model in part __b.__? Does this surprise you? Why or why not?

The LogisticRegression model trained in part_c() and shown in the top right plot was trained with an additional feature "gender" compared to the model in part **b**. The model equation in part_c() is g(0.048*I + 0.075*C + 0.135*G - 6.618), where G is the gender feature.

The decision boundary of this model is different from the one in part **b**. It can be seen from the top right plot that the decision boundary for this model is a linear function that separates the data into two classes based on their features. The fairness metrics for this model are also different from those of the model in part **b**. The PPV and NPV for each group are the same as those in part **b**, but the FPR and FNR for Group 1 are higher than those in part **b**, while the FNR for Group 0 is slightly higher.

This difference in fairness metrics is not surprising, as adding the gender feature may introduce bias into the model. In this case, the model may be more likely to predict that men are less likely to be approved for a loan than women, regardless of their income or credit score. This can be seen in the higher FPR and FNR for Group 1, which consists mostly of men.

Overall, the addition of the gender feature does not seem to improve the fairness of the model, as the fairness metrics are worse than those in part **b**. This suggests that further analysis and modifications may be necessary to ensure that the model is fair and unbiased.

d. Look at the code for both LogisticRegression models trained in `part_d()` and visualized in the bottom two plots of `free_response/5a.png`.

*** What is different about how each of these two models were trained?**

*** If you were applying for a loan and were a member of Group 0, would you rather be classified by the part __d.__ "G=0" classifier or the classifier from part __b.__? Why?**

*** If you were applying for a loan and were a member of Group 1, would you rather be classified by the part __d.__ "G=1" classifier or the classifier from part __b.__? Why?**
The two LogisticRegression models trained in part_d() and visualized in the bottom two plots are trained on subsets of the data where the gender is either 0 or 1. The G = 0 model is trained only on data where gender is 0, and the G = 1 model is trained only on data where gender is 1.

If you were applying for a loan and were a member of Group 0, you would rather be classified by part **d.** "G = 0" classifier, because it has a higher PPV and NPV than the classifier from part **b**. This means that if the classifier predicts that you are likely to be approved for a loan, there is a higher chance that you will actually be approved, and if it predicts that you are unlikely to be approved, there is a higher chance that you will actually be denied.

If you were applying for a loan and were a member of Group 1, you would rather be classified by the classifier from part **b**, because the G = 1 model from part **d.** has a perfect PPV and NPV, but this may be due to overfitting on the small subset of data where gender is 1. The classifier from part **b** has lower FPR and FNR for Group 1, which suggests that it is more reliable in predicting loan approval for this group.

e. The [US Equal Credit Opportunity Act (ECOA)](https://www.justice.gov/crt/equal-credit-opportunity-act-3) forbids "credit discrimination on the basis of race, color, religion, national origin, sex, marital status, age, or whether you receive income from a public assistance program." Suppose a bank wants to use machine learning to decide whether to approve or reject a loan application, but also wants to comply with the ECOA. What are __two__ additional challenges that might be introduced if your ML system needs to be fair with respect to eight protected attributes instead of just one?
If a bank wants to use machine learning to decide whether to approve or reject a loan application, while also complying with the ECOA to avoid credit discrimination based on multiple protected attributes, two additional challenges that may arise are:

1. Intersectional fairness: The interactions between protected attributes can lead to unfair outcomes. For example, if a bank's model unfairly approves loans for white married individuals who have a higher income and lives in certain neighborhoods, the model may still be discriminatory towards Black and Hispanic married individuals with the same income levels and living in the same neighborhoods. In this case, the model would fail to capture the intersectional identities of individuals, which can lead to biases and discrimination.
2. Data scarcity and quality: With multiple protected attributes, it can be challenging to obtain sufficient data for each subgroup to ensure that the model has enough representation from all groups. This data scarcity can cause models to be biased towards

the majority group, resulting in poor performance for underrepresented groups. Furthermore, data quality can also become an issue as data may be incomplete, inaccurate, or out of date, especially for smaller subgroups, which can result in discriminatory outcomes.